

```
print("VARIABLES")
```

```
name = "Sowmiya"
print(name)
```

```
a = 10
b = 20
a, b = b, a
print("a:", a, "b:", b)
```

```
x, y, z = 1, 2, 3
print("Sum:", x + y + z)
```

```
val = 100
val = 200
print(val)
```

```
num = 50
del num
```

```
print("DATA TYPES")
```

```
i = 10
f = 3.14
s = "Hello"
b = True

print(type(i), type(f), type(s), type(b))
```

```
<class 'int'> <class 'float'> <class 'str'> <class 'bool'>
```

```
print(type(3.14), type("Hello"), type(True), type(0))
```

<class 'float'> <class 'str'> <class 'bool'> <class 'int'>

```
x = 9
print(isinstance(x, int))
```

True

```
print(type(None))
```

```
<class 'NoneType'>
```

[illegible]

```
print("TYPE CONVERSION")
```

TYPE CONVERSION

```
num = int("123") + 7
print(num)
```

```
130
```

```
print(int(9.99))
```

```
9
```

```
print(bool(0), bool(1))
```

```
False True
```

```
print(int(True), int(False))
```

```
1 0
```

```
user_input = float(input("Enter a number: "))  
print(user_input * 2)
```

```
Enter a number: 23  
46.0
```

```
try:  
    print(int("hello"))  
except:  
    print("Error: Cannot convert 'hello' to integer")
```

```
Error: Cannot convert 'hello' to integer
```

```
print("STRING MANIPULATION")
```

```
STRING MANIPULATION
```

```
text = "Hello, World!"  
print(text.upper())  
print(text.lower())
```

```
HELLO, WORLD!  
hello, world!
```

```
print(len("Python Programming"))
```

```
18
```

```
print("Hello, World!"[7:12])
```

```
World
```

```
print("abcdef"[::-1])
```

```
fedcba
```

```
print("nan" in "banana")
```

```
True
```

```
print("This is a bad day".replace("bad", "good"))
```

```
This is a good day
```

```
print(" hello ".strip())
```

```
hello
```

```
print("apple,banana,cherry".split(","))
```

```
['apple', 'banana', 'cherry']
```

```
print("banana".count("a"))
```

```
3
```

```
print("Python123".isalnum())
```

```
True
```

```
print("STRING FORMATTING")
```

```
STRING FORMATTING
```

```
name = "John"
age = 25
print(f"My name is {name} and I am {age} years old")
```

```
My name is John and I am 25 years old
```

```
print("The price is {:.2f} dollars".format(49.99))
```

```
The price is 49.99 dollars
```

```
print(f"{3.14159:.2f}")
```

```
3.14
```

```
print("{:03}".format(7))
```

```
007
```

```
print(f"5 x 3 = {5*3}")
```

```
5 x 3 = 15
```

```
names = ["Alice", "Bob", "Charlie"]
scores = [85, 90, 95]
```

```
for n, s in zip(names, scores):
    print(f"{n:<10} {s:<5}")
```

```
Alice      85
Bob        90
Charlie    95
```

```
print("PYTHON LISTS")
```

```
PYTHON LISTS
```

```
fruits = ["apple", "banana", "cherry", "mango", "grape"]
print(fruits)
```

```
['apple', 'banana', 'cherry', 'mango', 'grape']
```

```
nums = [10, 20, 30, 40, 50]
print(nums[0], nums[-1])
```

```
10 50
```

```
nums.append(60)
print(nums)
```

```
[10, 20, 30, 40, 50, 60]
```

```
nums.insert(2, 99)
print(nums)
```

```
[10, 20, 99, 30, 40, 50, 60]
```

```
fruits.remove("banana")
print(fruits)
```

```
['apple', 'cherry', 'mango', 'grape']
```

```
lst = [5, 2, 8, 1, 9]
print(sorted(lst))
print(sorted(lst, reverse=True))
```

```
[1, 2, 5, 8, 9]
[9, 8, 5, 2, 1]
```

```
lst2 = [3, 7, 1, 9, 4]
print(len(lst2), max(lst2), min(lst2))
```

```
5 9 1
```

```
print([0, 1, 2, 3, 4, 5, 6][2:5])
```

```
[2, 3, 4]
```

```
copy_list = lst2.copy()
copy_list.append(100)
print("Original:", lst2)
print("Copy:", copy_list)
```

```
Original: [3, 7, 1, 9, 4]
Copy: [3, 7, 1, 9, 4, 100]
```

```
merged = [1, 2, 3] + [4, 5, 6]
print(merged)
```

```
[1, 2, 3, 4, 5, 6]
```